

Dexter Fernandes

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https://github.com/Dexter-Fernandes

AI Engineer

AI Engineer with 3+ years shipping production Python systems, focused on retrieval-augmented generation and evaluation for financial-document question answering. Built an end-to-end RAG evaluation harness with a fine-tuned, self-hosted domain-adapted reranker and a stateful agent using tool use and multi-step reasoning, attributing failures across retrieval, generation, and grounding. Background in production computer vision and time-series ML, with SQL, pandas, and messy real-world data.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built a **RAG evaluation harness** for financial-document question answering to diagnose whether failures stem from retrieval, generation, or grounding, rather than reporting a single opaque accuracy figure.
- Built a **hybrid-retrieval** pipeline combining **BM25**, **FAISS** dense embeddings and **reciprocal rank fusion** with **cross-encoder reranking**, using **recursive character chunking** and three isolable retrieval modes (closed-book, oracle-context, dense) to separate retrieval quality from generation quality.
- Fine-tuned a **bge-reranker-base cross-encoder** with **LoRA (PEFT)** and hard-negative mining for financial-document domain adaptation, **self-hosted** behind a containerised inference API, and benchmarked it against the pretrained model on MRR, nDCG, evidence hit rate, grounding, and downstream answer accuracy, improving retrieval ranking over the pretrained baseline.
- Built a stateful **LangGraph** agent that orchestrates retrieval, correctness, and grounding tools through conditional workflows with multi-step reasoning and structured **Pydantic** outputs, benchmarking diagnostic accuracy, latency, and tool usage against a deterministic baseline.
- Designed a dual-track evaluation layer pairing deterministic checks (exact/normalised match, scale-aware numeric matching) with **LLM-as-judge** scoring, joined against retrieval metrics to attribute wrong answers to retrieval, generation, or grounding failure.
- Benchmarked the pipeline against **Snowflake Cortex Search** on identical chunks and metrics, and exposed runs, results, and failure diagnostics through typed MCP (Model Context Protocol) tools for programmatic and human-readable inspection.

WORK EXPERIENCE

Independent Projects & Upskilling

Sep 2024 - Present

Self-directed

- Self-directed AI engineering focused on RAG, agent workflows, and LLM evaluation, producing the FinanceBench RAG evaluation harness as the primary output.
- Completed Nvidia "Building RAG Agents with LLMs" (Jul 2026), covering RAG, agents, and evaluation with LangChain and LangSmith.

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the **end-to-end computer vision pipeline** for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Built a separate drilling-activity classification workstream over 91 WITS-system sensor channels, using **gradient boosting**, **random forests**, and **MultiRocket** for multivariate time-series classification to support operational decisions from real-time rig telemetry.

- Built **multimodal data ingestion**, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, with PostgreSQL for structured pipeline and metadata storage, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Presented model performance, failure modes, and **deployment trade-offs** to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.

LivSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries and John Deere.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.

Bachelor of Technology in Electrical and Electronics Engineering

Vit Vellore Institute of Technology • Vellore, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Nvidia

Jul 2026

Deep Learning Specialisation

deeplearning.ai

Jul 2019

SKILLS

- **AI-Assisted Development:** Claude Code, ChatGPT Codex
- **RAG and Retrieval:** document-grounded QA, retrieval metrics, RAG evaluation, RAG pipelines, dense retrieval, hybrid retrieval, FAISS vector store, ChromaDB, BM25, reciprocal rank fusion, cross-encoder reranking
- **Agent Engineering and Evaluation:** LLM-as-judge, agent orchestration, tool orchestration, stateful agent workflows, MCP, prompt engineering, evidence matching, citation checks, hallucination detection, error analysis, structured outputs, FastAPI, multi-step reasoning
- **Experimentation and Reliability:** failure-mode analysis, benchmarking, regression checks, hypothesis testing, ablation studies, post-deployment validation, production monitoring
- **Python and ML Engineering:** Python, Pandas, NumPy, scikit-learn, PyTorch, Apache Airflow, Optuna, LangChain, LangSmith, PyDantic, FastAPI, Snowflake, LangGraph, fine-tuning, LoRA, computer vision, object detection, gradient boosting, random forests, MultiRocket

- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux, Microsoft Azure, SQL, PostgreSQL, self-hosted model serving, containerised inference